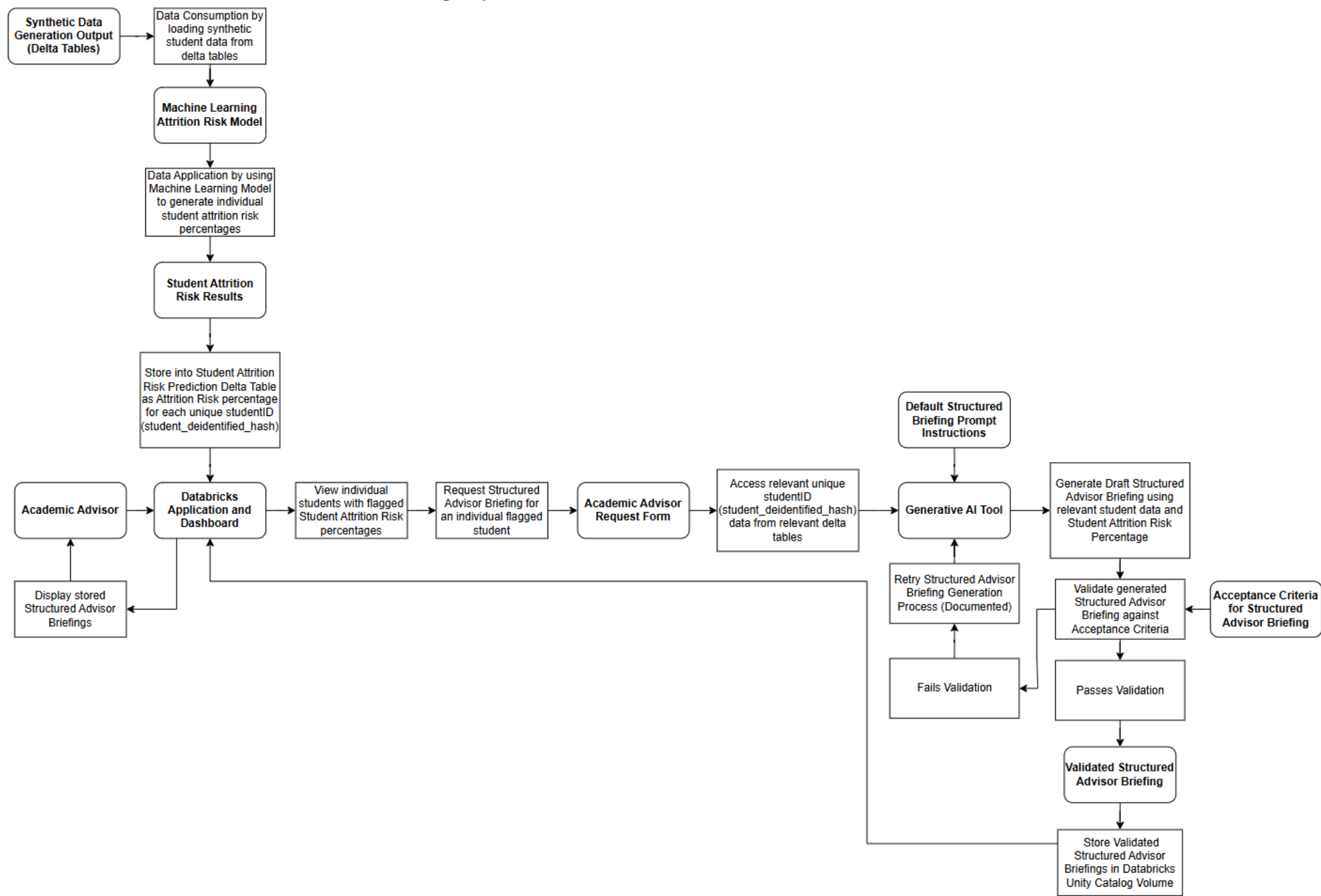


Student Attrition Risk and Intervention Briefing Physical:



T115 Student Risk & Intervention Briefing Documentation

As of 22/07/2026

Built:

- Synthetic Data Generation process as input required for the full Student Attrition Risk & Intervention Briefing use case has been fully built.
- The physical solution design of synthetic data generation and the corresponding documentation has been completed to provide further context.

Intended to be built:

- **Machine Learning Model** to predict Student Attrition Risk (Data Consumption of the synthetic data generation input).
- **Databricks Application and Dashboard** (Data Application of the Machine Learning Model results for Student Attrition Risk Percentage, frontend dashboard interface focuses on design and user experience).
- **Databricks Application Generative AI Tool**. Backend AI API utilizing prompt engineered prompts in addition to synthetic data generated in delta tables regarding individual unique students, relevant features for the machine learning model, and student attrition risk percentage to generate main output of structured advisor briefings.

Inputs:

- **Synthetic Data Generation** is used as input for the Machine Learning Model to predict a student attrition risk percentage for each student.
- **Default Structured Briefing Prompts** act as a guideline for the initial prompt to generate a structured academic advisor briefing for a student at risk of attrition.
- **Acceptance Criteria for Structured Advisor Briefing** details the necessary acceptance criteria to accept a generated structured advisor, this is provided in the initial default prompt to help prompt engineer the highest likelihood of successful generation and reduce usage of failed generation and unnecessary API token usage.

Components and actual tools:

- **Databricks Delta tables** store the synthetic student data and individual student attrition risk prediction results.
- **Machine Learning Classification Model** developed in Python and managed through MLflow generates a student attrition risk percentage for each student.
- **Databricks Application and dashboard** allow academic advisors to view students at risk of student attrition, select a student at risk and request a briefing.
- **Generative AI API tool** (OpenAI) generates and revises the Structured Advisor Briefing.
- **Unity Catalog Volume** stores validated briefing documents so they can be reopened through the application.

How the solution works:

- The Machine Learning Model acts as a component for data consumption, consuming the synthetic student data generated from the synthetic data generation process to generate an individual student attrition risk percentage.
- Prediction is linked to relevant student_deidentified_hash column and stored in a Student Attrition Risk Prediction Delta Table.
- Databricks Application displays students at risk of attrition and allows academic advisors to request briefing for a selected student that is at risk.
- Databricks Application retrieves relevant information regarding student's attrition risk result and other associated relevant information (student information and relevant features for the Machine Learning model that affected the student attrition risk percentage variable). This information is sent to the Generative AI Tool that uses this information in the initial instructional prompt.
- Generated briefing is validated against Acceptance Criteria, and takes two routes depending on pass or failure of the generated briefing.
- If **passes** Acceptance Criteria validation, the briefing is stored in Unity Catalog Volume for the backend of the application and displayed in the frontend dashboard.
- If **fails** the Acceptance Criteria validation, then the application allows one attempt feeding failure information to generate another briefing through the Generative AI Tool. If second validation check fails, the application displays an error and does not store a briefing.

Transformations:

- Student Data Generation process that is stored in saved delta tables is transformed into an individual student attrition risk percentage prediction in the Machine Learning model (data consumption into transformation).
- Selected student attrition risk result, relevant student information, and relevant feature to predict student attrition risk percentage is used with default briefing instructions for the Generative AI API Tool to prompt. This transforms stored delta table information into an engineered prompt for the Generative AI Model.
- Generative AI transforms supplied student data, student attrition risk result, and initial engineered prompt into a draft Structured Advisor Briefing.

Outputs:

- Individual Student Attrition Risk Percentage as an output of the Machine Learning Model for Student Attrition Risk.
- Student Attrition Risk Flag applicable to risk threshold of student attrition risk percentage (0.5 prediction threshold or 50% binary student attrition risk flag).
- Validated Structured Advisor Briefing document that has passed the defined acceptance criteria.
- Stored Structured Advisor Briefing Document in Unity Catalogue Volume so it can be accessed through the Databricks application for an academic advisor. Non-validated Structured Advisor Briefing documents are not saved in any database system (Unity Catalog Volume).

Validation:

- Validate the primary output document of a generated Structured Advisor Briefing against defined acceptance criteria documentation before it is treated as a final output.
- Acceptance criteria documentation includes relevant information necessary for generation of a Structured Advisor Briefing, in addition to ensuring relevant informational sections, AI interpretations of results, and appropriate language for academic advisors is presented.
- Failed validation result identifies acceptance criteria were not met and produces correction feedback to be used in second attempt of the Generative AI Tool generating a Structured Advisor Briefing.
- The application permits one Structured Advisor Briefing regeneration attempt using a revised prompt containing the Validation Feedback and failed acceptance criteria.
- On second attempt at generating a Structured Advisor Briefing with the Generative AI Tool, if this attempt fails an error is displayed and the briefing is not stored as a validated output.
- Retry Structured Advisor Briefing Generation process shown in the physical model detailed above can be summarized as: a revised prompt using the failed acceptance criteria and validation feedback to generate the briefing one additional time on failure, if the regenerated briefing also fails validation, an error message is displayed on the dashboard.

Generative AI Process:

- Application combines selected student's relevant information, student attrition risk percentage, and default briefing instructions into a targeted prompt.
- Generative AI Tool transforms this information into a draft Structured Advisor Briefing, ready to be checked against acceptance criteria.
- Briefing describes student factors through AI interpretation that **may be** associated with the student attrition risk result. These are clearly presented to academic advisors as AI interpretations of the result in the Structured Advisor Briefing, not as confirmed causation of information resulting in student attrition risk percentage.
- The validation process returns a pass or fail result and identifies any failed acceptance criteria.
- When validation fails, original instructions and validation feedback are combined into a regenerated retry prompt. The Generative AI tool uses this revised prompt to attempt creating the Structured Advisor Briefing for the second time.

End-to-end flow

- The Academic Advisor opens the Databricks Application.
- The application retrieves Student Attrition Risk Percentages from the Student Attrition Risk Prediction Delta Table.
- The dashboard displays students identified as being at risk from the Student Attrition Risk Percentages.

- The academic advisor selects an individual student and clicks to request a Structured Advisor Briefing.
- The application retrieves relevant information from the saved delta tables (relevant student information, student's attrition risk percentage, features highly predictive of the student attrition risk percentage to be provided in the Structured Advisor Briefing).
- Databricks Application calls the Generative AI API to use this relevant information along with pre-made default briefing prompt instructions in a prompt to generate a draft Structured Advisor Briefing.
- Draft Structured Advisor Briefing is validated against defined acceptance criteria of a Structured Advisor Briefing.
- If validation fails, the Retry Structured Advisor Briefing Generation Process uses the failed acceptance criteria and validation feedback to create a revised prompt and generate the briefing one additional time. If the regenerated briefing also fails validation, the application displays an error and no briefing is stored.
- If validation passes, draft briefing is generated as a final briefing document. This document is stored in a Unity Catalog Volume and is now visible for academic advisors.
- Academic Advisor is now able to see the finalised Structured Advisor Briefing in the Databricks application dashboard.